

DW_fp_cmp_DG

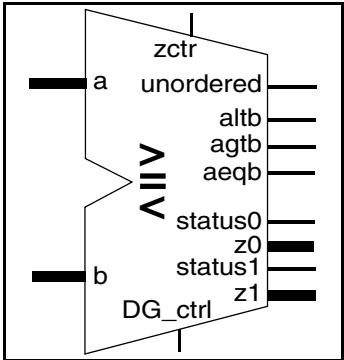
Floating-Point Comparator with Datapath Gating

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Features and Benefits

- Has the same functionality as DW_fp_cmp when the component is in normal operation
- Consumes less dynamic power than DW_fp_cmp when disabled (inputs are active, but the output is not being used)
- The precision format is parameterizable for either IEEE single, double precision, or a user-defined custom format
- Exponents can range from 3 to 31 bits
- Significand and fractional part of the floating-point number can range from 2 to 256 bits
- DesignWare datapath generators are employed for better power and QoR

Revision History



Description

DW_fp_cmp_DG is a floating-point module that compares two floating-point numbers at inputs *a* and *b*. The format of these numbers is defined by the number of bits in the exponent (*exp_width*) and the number of bits in the significand (*sig_width*). Only one of the four outputs: *agtb*, *altb*, *aeqb* or *unordered* is set as a result of this operation. The input *zctr* is used to select the function that generates the *z0* and *z1* outputs. Also, a control input (*DG_ctrl*) can disable the component to reduce dynamic power consumption when the component is not in use and inputs are still active. DW_fp_cmp_DG also produces *status0* and *status1* with information about the *z0* and *z1* outputs. For detailed behavior, see [Table 1-5](#) on page 4 and [Table 1-6](#) on page 5.

[Table 1-1](#) contains pin descriptions and configuration parameters are listed in [Table 1-2](#) on page 2.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	<i>sig_width</i> + <i>exp_width</i> + 1 bits	Input	Floating-point number
b	<i>sig_width</i> + <i>exp_width</i> + 1 bits	Input	Floating-point number
altb	1 bit	Output	High when <i>a</i> is less than <i>b</i>
agtb	1 bit	Output	High when <i>a</i> is greater than <i>b</i>
aeqb	1 bit	Output	High when <i>a</i> is equal to <i>b</i>
unordered	1 bit	Output	High when one of the inputs is NaN and <i>ieee_compliance</i> = 1

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
z0	$sig_width + exp_width + 1$ bits	Output	- When <code>zctr</code> is 0 or open, $z0 = \text{Min}(a,b)$ - When <code>zctr</code> is 1, $z0 = \text{Max}(a,b)$
z1	$sig_width + exp_width + 1$ bits	Output	- When <code>zctr</code> is 0 or open, $z1 = \text{Max}(a,b)$ - When <code>zctr</code> is 1, $z1 = \text{Min}(a,b)$
status0 ^a	8 bits	Output	- status0[0:6]: Status flags corresponding to <code>z0</code>; for details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> - status0[7]^b: Set when operand <code>a</code> is passed to output <code>z0</code>
status1 ^a	8 bits	Output	- status1[0:6]: Status flags corresponding to <code>z1</code>; for details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> - status1[7]^b: Set when operand <code>a</code> is passed to output <code>z1</code>
zctr	1 bit	Input	Determines value passed to <code>z0</code> and <code>z1</code>
DG_ctrl	1 bit	Input	Datapath gating control 0 = Component is disabled 1 = Normal component operation See “Datapath Gating Control with DG_ctrl” on page 6.

a. DW_fp_cmp_DG does not use bit 3 (the Tiny bit) or bit 4 (the Huge bit) of `status0` and `status1` because rounding is not performed

b. Bit 7 helps to trace the input that has the $\text{Max}(a,b)$ or $\text{Min}(a,b)$ value, based on the functionality of the `z0` and `z1` outputs (as determined by `zctr`).

For example, when `zctr`=0, $z0=\text{Min}(a,b)$, $z1=\text{Max}(a,b)$. So, when `a`=3 and `b`=4:

$z0=\text{Min}(3,4)=3$ and `status0`[7]=1

$z1=\text{Max}(3,4)=4$, and `status1`[7]=0

Looking at `status0`[7] and `status1`[7], you can see that the minimum input value (`z0` output) is coming from input `a` (because `status0`[7] is set), and the maximum input value is coming from input `b`.

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 253 bits Default: 23	Word length of fraction field of floating-point numbers <code>a</code> , <code>b</code> , <code>z0</code> , and <code>z1</code> .
exp_width	3 to 31 bits Default: 8	Word length of biased exponent of floating-point numbers <code>a</code> , <code>b</code> , <code>z0</code> , and <code>z1</code> .
ieee_compliance	0 or 1 Default: 0	0: NaNs and denormals are unsupported. NaNs are considered infinities and denormals are considered zeros. 1: The generated architecture is fully compliant with IEEE 754 standard, including the use of denormals and NaNs.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Datapath gating close to the main inputs	■ DesignWare (P-2019.03 and later) ■ DesignWare-LP (before P-2019.03)

Table 1-4 Simulation Models

Model	Function
DW02.DW_FP_CMP_DG_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_cmp_DG_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_fp_cmp_DG.v	Verilog simulation model source code

The parameter *ieee_compliance* controls the functionality of this component.

- When *ieee_compliance* = 0, the DW_fp_cmp_DG component does not support NaNs and denormals. The *aeqb*, *altb*, and *agtb* outputs are generated considering denormal inputs as zeros, and the *z0* and *z1* outputs pass the values of inputs *a* and *b* without forcing them to zeros. Also, the *unordered* output is always zero in this configuration.
- When the *ieee_compliance* = 1, the component is fully compliant with the IEEE 754 standard and therefore operates with NaNs and denormals. The *unordered* output is set to one when a NaN value is applied as input.

For detailed behavior when *ieee_compliance* = 0, see [Table 1-5](#) on page 4.

For detailed behavior when *ieee_compliance* = 1, see [Table 1-6](#) on page 5.

Table 1-5 DW_fp_cmp_DG Behavior When *sig_width* = 10, *exp_width* = 5, and *ieee_compliance* = 0

a	b	zctr	z0	z1	status0	status1	altb	agtb	aeqb	unordered
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal: zero)	0	0_00000_0110000000 (min(a,b)= b)	0_00011_0000110110 (max(a,b)= a)	00000001	10000000	0	1	0	0
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal: zero)	1	0_00011_0000110110 (max(a,b)= a)	0_00000_0110000000 (min(a,b)= b)	10000000	00000001	0	1	0	0
0_00000_0000110110 (denormal: zero)	1_00000_0110000000 (denormal: zero)	0	0_00000_0000110110 (a input)	1_00000_0110000000 (b input)	10000001	00000001	0	0	1	0
0_00000_0000110110 (denormal: zero)	1_00000_0110000000 (denormal: zero)	1	1_00000_0110000000 (b input)	0_00000_0000110110 (a input)	00000001	10000001	0	0	1	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	0	0_00000_0000000000 (a input)	1_00000_0000000000 (b input)	10000001	00000001	0	0	1	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	1	1_00000_0000000000 (b input)	0_00000_0000000000 (a input)	00000001	10000001	0	0	1	0
1_11111_0101100111 (NaN: -infinity)	0_10111_0010011010 (normal number)	0	1_11111_0101100111 (min(a,b)= a)	0_10111_0010011010 (max(a,b)= b)	10000010	00000000	1	0	0	0
1_11111_0101100111 (NaN: -infinity)	0_10111_0010011010 (normal number)	1	0_10111_0010011010 (max(a,b)= b)	1_11111_0101100111 (min(a,b)= a)	00000000	10000010	1	0	0	0
0_11111_0101100111 (NaN: +infinity)	1_11111_0010011010 (NaN: -infinity)	0	1_11111_0010011010 (min(a,b) = b)	0_11111_0101100111 (max(a,b) = a)	00000010	10000010	0	1	0	0
0_11111_0101100111 (NaN: +infinity)	1_11111_0010011010 (NaN: -infinity)	1	0_11111_0101100111 (max(a,b) = a)	1_11111_0010011010 (min(a,b) = b)	10000010	00000010	0	1	0	0
0_11111_0101100111 (NaN: +infinity)	0_11111_0010011010 (NaN: +infinity)	0	0_11111_0101100111 (a input)	0_11111_0010011010 (b input)	10000010	00000010	0	0	1	0
0_11111_0101100111 (NaN: +infinity)	0_11111_0010011010 (NaN: +infinity)	1	0_11111_0010011010 (b input)	0_11111_0101100111 (a input)	00000010	10000010	0	0	1	0

Table 1-6 DW_fp_cmp_DG Behavior When *sig_width* = 10, *exp_width* = 5, and *ieee_compliance* = 1

a	b	zctr	z0	z1	status0	status1	altb	agtb	aeqb	unordered
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal)	0	0_00000_0110000000 (min(a,b)= b)	0_00011_0000110110 (max(a,b)= a)	00000000	10000000	0	1	0	0
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal)	1	0_00011_0000110110 (max(a,b)= a)	0_00000_0110000000 (min(a,b)= b)	10000000	00000000	0	1	0	0
0_00000_0000110110 (denormal)	1_00000_0110000000 (denormal)	0	1_00000_0110000000 (min(a,b) = b)	0_00000_0000110110 (max(a,b) = a)	00000000	10000000	0	1	0	0
0_00000_0000110110 (denormal)	1_00000_0110000000 (denormal)	1	0_00000_0000110110 (max(a,b) = a)	1_00000_0110000000 (min(a,b) = b)	10000000	00000000	0	1	0	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	0	0_00000_0000000000 (a input)	1_00000_0000000000 (b input)	10000001	00000001	0	0	1	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	1	1_00000_0000000000 (b input)	0_00000_0000000000 (a input)	00000001	10000001	0	0	1	0
1_11111_0101100111 (NaN)	0_10111_0010011010 (normal number)	0	1_11111_0101100111 (a input)	0_10111_0010011010 (b input)	10000100	00000000	0	0	0	1
1_11111_0101100111 (NaN)	0_10111_0010011010 (normal number)	1	1_11111_0101100111 (a input)	0_10111_0010011010 (b input)	10000100	00000000	0	0	0	1
0_11111_0101100111 (NaN)	1_11111_0010011010 (NaN)	0	0_11111_0101100111 (a input)	1_11111_0010011010 (b input)	10000100	00000100	0	0	0	1
0_11111_0101100111 (NaN)	1_11111_0010011010 (NaN)	1	0_11111_0101100111 (a input)	1_11111_0010011010 (b input)	10000100	00000100	0	0	0	1
0_11111_0101100111 (NaN)	0_11111_0010011010 (NaN)	0	0_11111_0101100111 (a input)	0_11111_0010011010 (b input)	10000100	00000100	0	0	0	1
0_11111_0101100111 (NaN)	0_11111_0010011010 (NaN)	1	0_11111_0101100111 (a input)	0_11111_0010011010 (b input)	10000100	00000100	0	0	0	1

Datapath Gating Control with DG_ctrl

For DW_fp_cmp_DG and other combinational components that have the datapath gating feature, the DG_ctrl port is provided to control datapath gating.

When DG_ctrl = 1, the component behaves as expected according to activity on the input ports.

When DG_ctrl = 0:

- The component is disabled and internal gates are totally or partially isolated to block propagation of switching activity inside the component. This makes the component less sensitive to switching activity on the main ports and reduces dynamic power consumption.
- Values at the output ports are not defined.
- Simulation models set 'X' values at the output ports.

Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Blocks User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_fp_cmp_DG_inst is
  generic (
    inst_sig_width : POSITIVE := 23;
    inst_exp_width : POSITIVE := 8;
    inst_ieee_compliance : INTEGER := 0
  );
  port (
    inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_b : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_zctr : in std_logic;
    inst_DG_ctrl : in std_logic;
    aeqb_inst : out std_logic;
    altb_inst : out std_logic;
    agtb_inst : out std_logic;
    unordered_inst : out std_logic;
    z0_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    z1_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    status0_inst : out std_logic_vector(7 downto 0);
    status1_inst : out std_logic_vector(7 downto 0)
  );
end DW_fp_cmp_DG_inst;

architecture inst of DW_fp_cmp_DG_inst is

begin

  -- Instance of DW_fp_cmp_DG
  U1 : DW_fp_cmp_DG
    generic map ( sig_width => inst_sig_width, exp_width => inst_exp_width,
    ieee_compliance => inst_ieee_compliance )
    port map ( a => inst_a, b => inst_b, zctr => inst_zctr, DG_ctrl => inst_DG_ctrl,
    aeqb => aeqb_inst, altb => altb_inst, agtb => agtb_inst, unordered => unordered_inst,
    z0 => z0_inst, z1 => z1_inst, status0 => status0_inst, status1 => status1_inst );

end inst;

```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_cmp_DG_inst( inst_a, inst_b, inst_zctr, inst_DG_ctrl, aeqb_inst,
    altb_inst, agtb_inst, unordered_inst, z0_inst, z1_inst,
    status0_inst, status1_inst );

parameter sig_width = 23;
parameter exp_width = 8;
parameter ieee_compliance = 0;

input [sig_width+exp_width : 0] inst_a;
input [sig_width+exp_width : 0] inst_b;
input inst_zctr;
input inst_DG_ctrl;
output aeqb_inst;
output altb_inst;
output agtb_inst;
output unordered_inst;
output [sig_width+exp_width : 0] z0_inst;
output [sig_width+exp_width : 0] z1_inst;
output [7 : 0] status0_inst;
output [7 : 0] status1_inst;

    // Instance of DW_fp_cmp_DG
    DW_fp_cmp_DG #(sig_width, exp_width, ieee_compliance)
        U1 ( .a(inst_a), .b(inst_b), .zctr(inst_zctr), .DG_ctrl(inst_DG_ctrl),
            .aeqb(aeqb_inst), .altb(altb_inst), .agtb(agtb_inst), .unordered(unordered_inst),
            .z0(z0_inst), .z1(z1_inst), .status0(status0_inst), .status1(status1_inst) );

endmodule
```


Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
December 2023	DWBB_202312.0	Updated date and version; no technical changes
October 2021	DWBB_202106.3	- Added footnote about bits 3 and 4 of <code>status0</code> and <code>status1</code> in Table 1-1 on page 1 - Clarified the releases noted in this table
July 2021	DWBB_202106.1	- Clarified behavior when <code>ieee_compliance</code> = 0 in Table 1-2 on page 2 and in text on page 3 - Clarified the purpose of <code>status0[7]</code> and <code>status1[7]</code> on page 2 - Added detailed behavior in Table 1-5 on page 4 and Table 1-6 on page 5
March 2019	DWBB_201903.0	- Added “Datapath Gating Control with DG_ctrl” on page 6 - Clarified some information about minPower - Added this Revision History table and the document links on this page

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